

8	A	GPE = KE $0.3 (9.81) (2) = 5.886 = \frac{1}{2} (0.3) v^2$ $v = 6.2641 \text{ m s}^{-1}$ $P_i = P_f$ $0.3 (6.2641) = (0.3 + 0.5) V$ $V = 2.3491 \text{ m s}^{-1}$ KE after collision = $\frac{1}{2} (0.8) 2.3491^2 = 2.20725 \text{ J}$ therefore EPE = $5.886 - 2.20725 = \frac{1}{2} k (0.20)^2$ $k = 184 \text{ N m}^{-1} \quad (3 \text{ s.f.})$
9	B	At B, resultant force towards centre of dome $F_{\text{net}} = mg \cos 48.2^\circ$